

Bachelor of Science in Computer Science and Information Technology

Shankhamul - 9 , Lalitpur, Nepal

Lab Report

Subject Code: CSC415



NAGARJUNA COLLEGE
of Information Technology
(Affiliated to Tribhuvan University)

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Subject: Software Project Management

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LAB NO. 1

Title: Creating an account in cloud based project management software and create a working team for a project.

Objective

- To create an account in cloud-based project management software.
- To understand the basic features of project management tools.
- To create and manage a working team for a project.
- To assign tasks and collaborate with team members online.

Theory

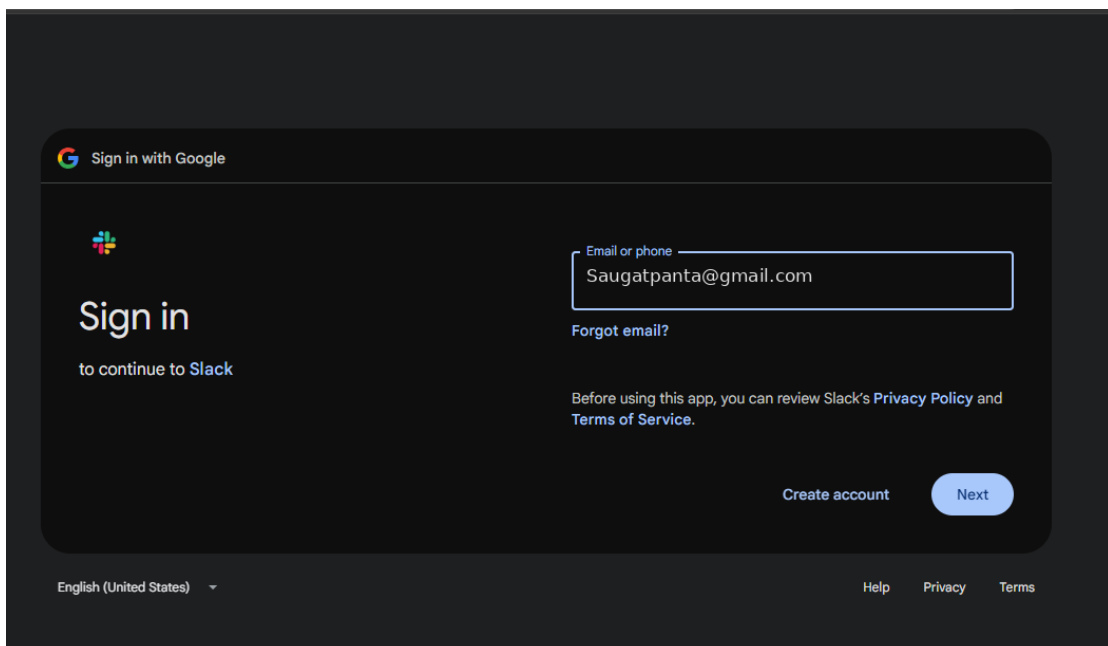
Cloud-based project management software is an online system used to manage projects, tasks, and team activities through the internet. It helps users create teams, communicate with members, assign tasks, and monitor project progress efficiently. These tools improve collaboration, productivity, and coordination among team members working on the same project.

Example: Slack

Procedure

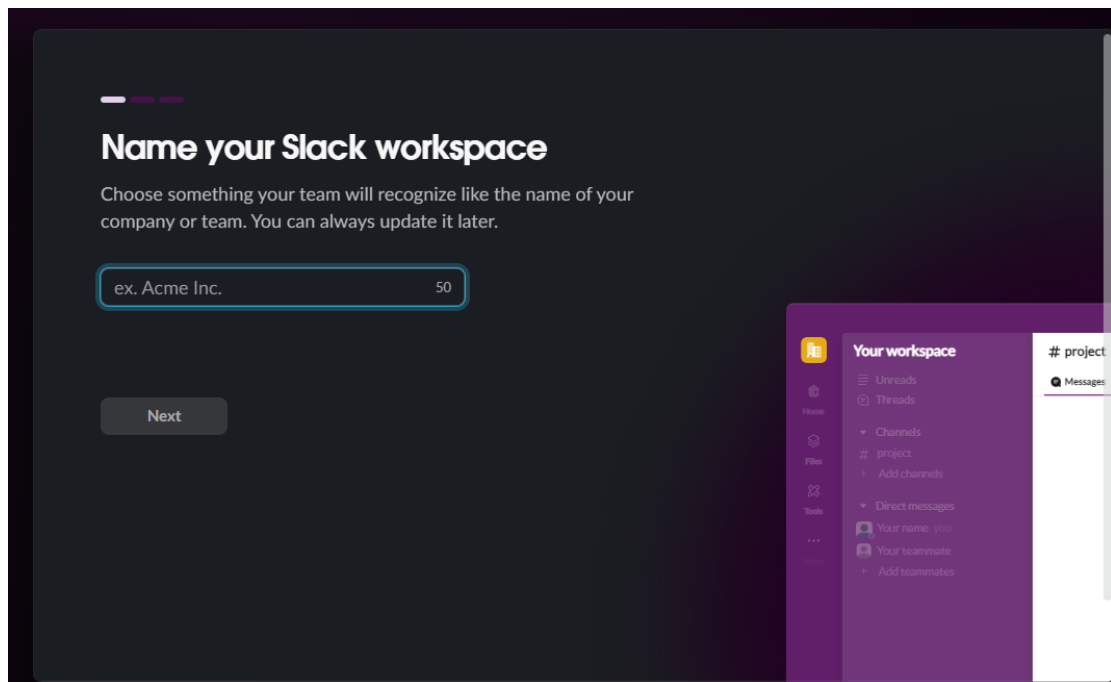
Step 1: Creating Slack Account

Opened the Slack website and entered the email address to create a new account.

A screenshot of the Slack sign-in page. The page has a dark background. At the top left, there is a 'Sign in with Google' button with the Google logo. Below this, on the left, is the Slack logo and the text 'Sign in to continue to Slack'. On the right, there is a text input field labeled 'Email or phone' containing the email 'Saugatpanta@gmail.com'. Below the input field is a link that says 'Forgot email?'. At the bottom right, there are two buttons: 'Create account' and 'Next'. At the very bottom of the page, there is a language selector showing 'English (United States)' and links for 'Help', 'Privacy', and 'Terms'.

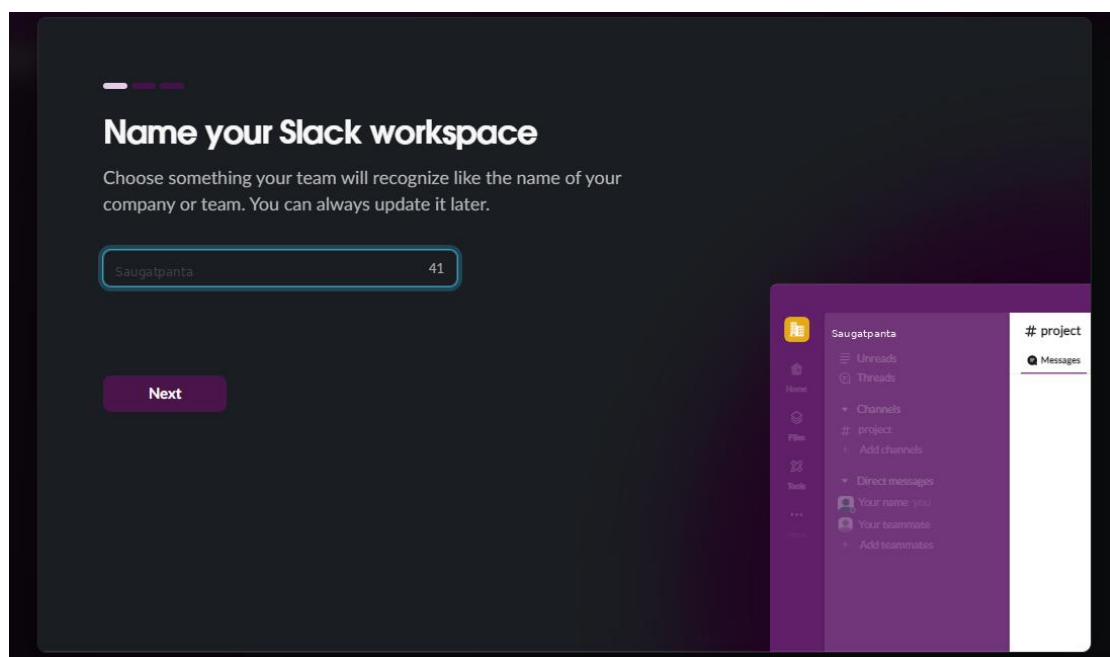
Step 2: Launching Workspace

After verification, a new Slack workspace was launched successfully.



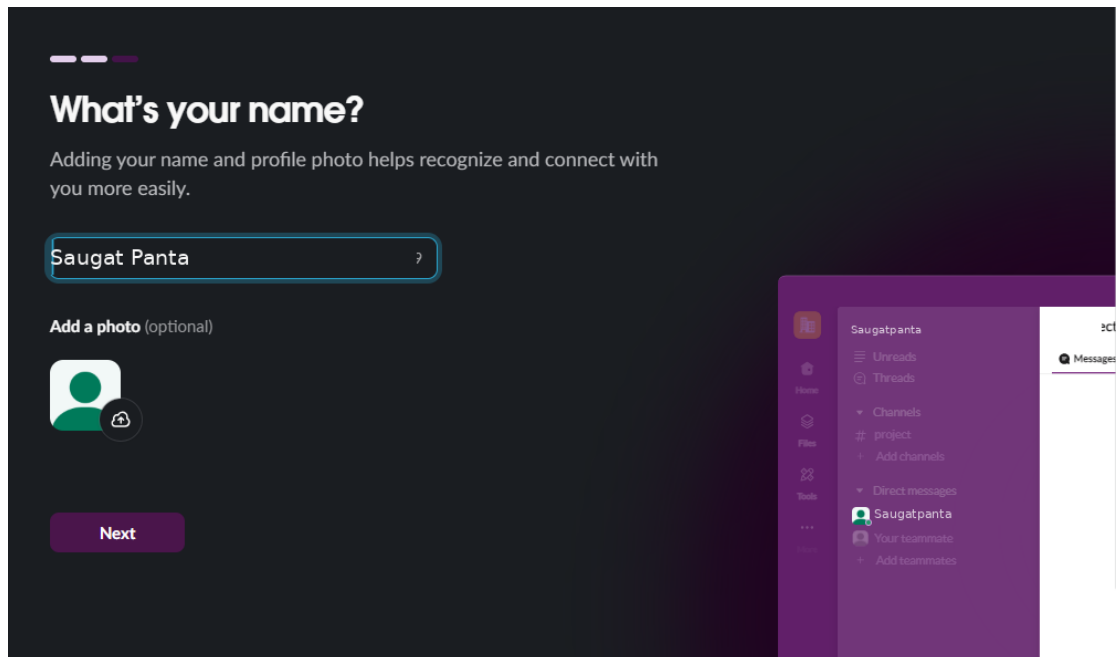
Step 3: Naming Workspace

Created a workspace with the name SPM_LAB.

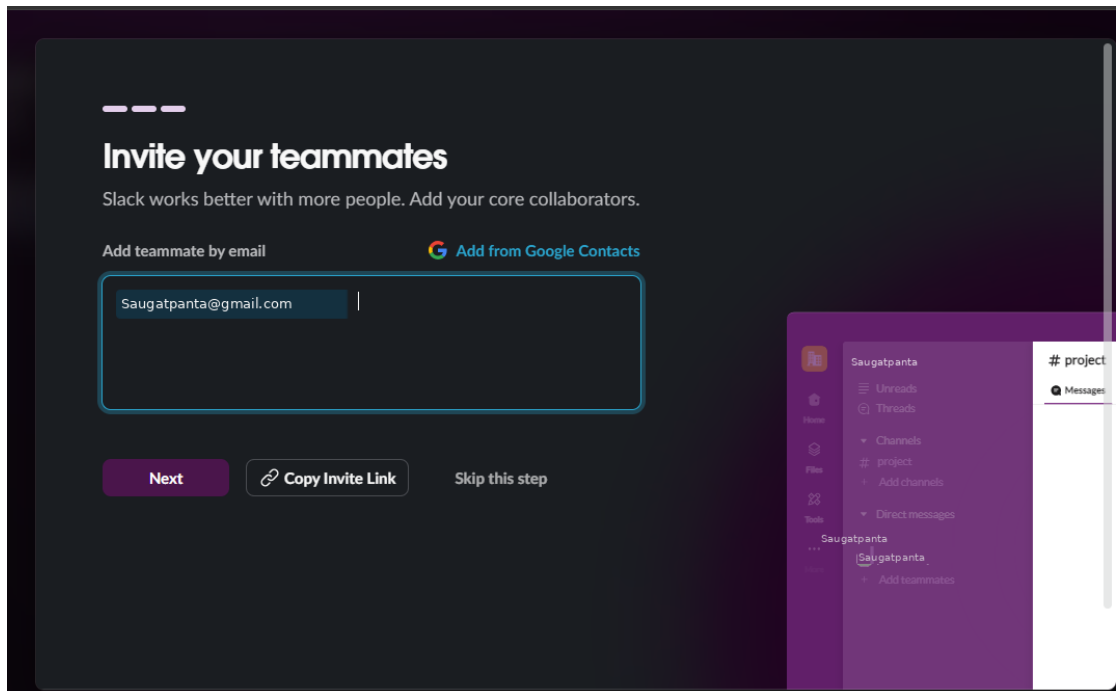


Step 4: Setting User Profile

Entered the user name Nishant Chaudhary and completed profile setup.



Step 5: Inviting Team Members



Invited team members using their email addresses to create a working team.

Result

A Slack account and workspace were successfully created, and team members were added to the project workspace.

Conclusion

Cloud-based project management software helps teams collaborate, communicate, and manage projects effectively from anywhere.

LAB NO. 2

Title: Managing relationships among tasks for Project Scheduling and creating a Gantt Chart.

Objective

- To understand task relationships in project scheduling.
- To manage dependent tasks in a project.
- To create a Gantt Chart for project planning and scheduling.

Theory

Project scheduling is the process of planning and organizing project tasks according to time and dependency. Task relationships define how one task is connected to another task in a project. These relationships help in maintaining proper workflow and project timing.

Common task relationships are:

- Finish to Start (FS)
- Start to Start (SS)
- Finish to Finish (FF)
- Start to Finish (SF)

A Gantt Chart is a graphical representation of project tasks along a timeline. It helps users track task duration, dependencies, and project progress effectively.

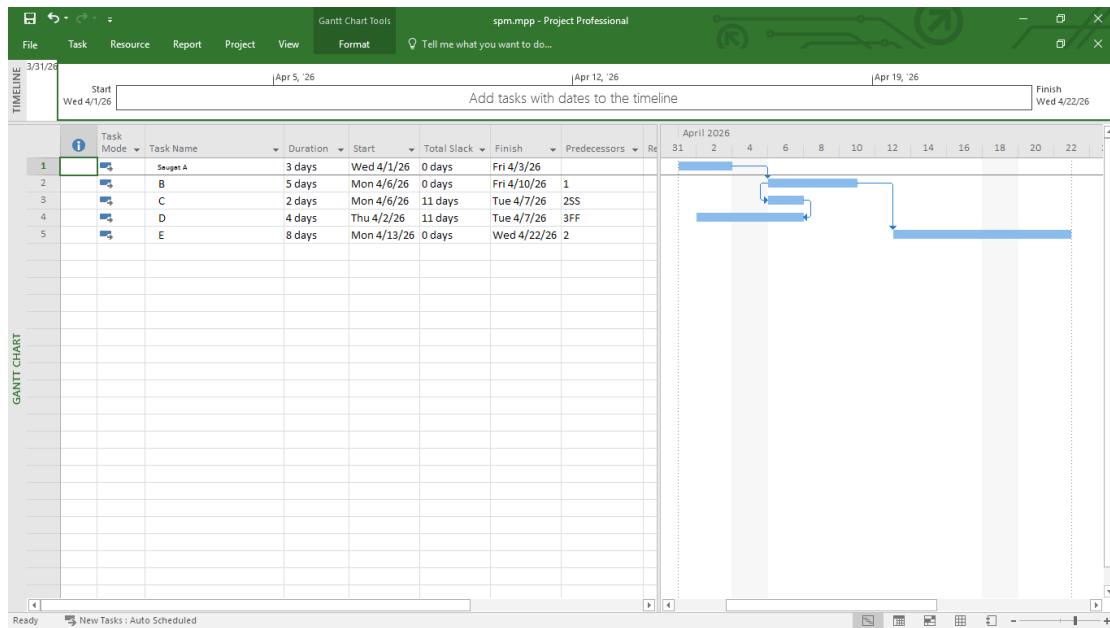
Procedure

Step 1: Open Project Management Software

Opened the project scheduling software and created a new project.

Step 2: Create Gantt Chart and Manage Task Relationships

Generated the Gantt Chart and added predecessor values to establish dependency relationships among tasks.



Result

Task relationships were successfully managed, and a Gantt Chart was created for project scheduling.

Conclusion

Managing task relationships and creating a Gantt Chart helps in effective project planning, scheduling, and monitoring project progress.

LAB NO. 3

Title: Creating a calendar for the project and adjusting duty shifts.

Objective

- To create a project calendar for scheduling activities.
- To adjust duty shifts and working time for project tasks.

Theory

A project calendar is used to schedule project activities, working days, and task timing. It helps project managers organize work according to available time and resources. Duty shift adjustment is the process of managing working hours, holidays, and employee shifts to maintain proper project workflow and productivity.

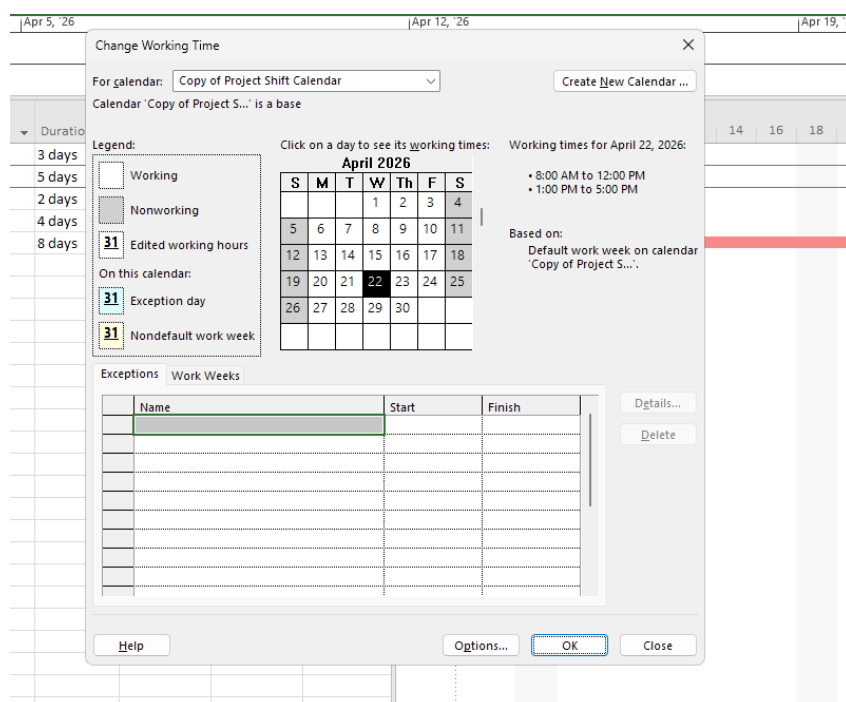
Project calendars help in:

- Managing working schedules
- Adjusting duty shifts
- Avoiding scheduling conflicts
- Improving project planning and resource utilization

Procedure

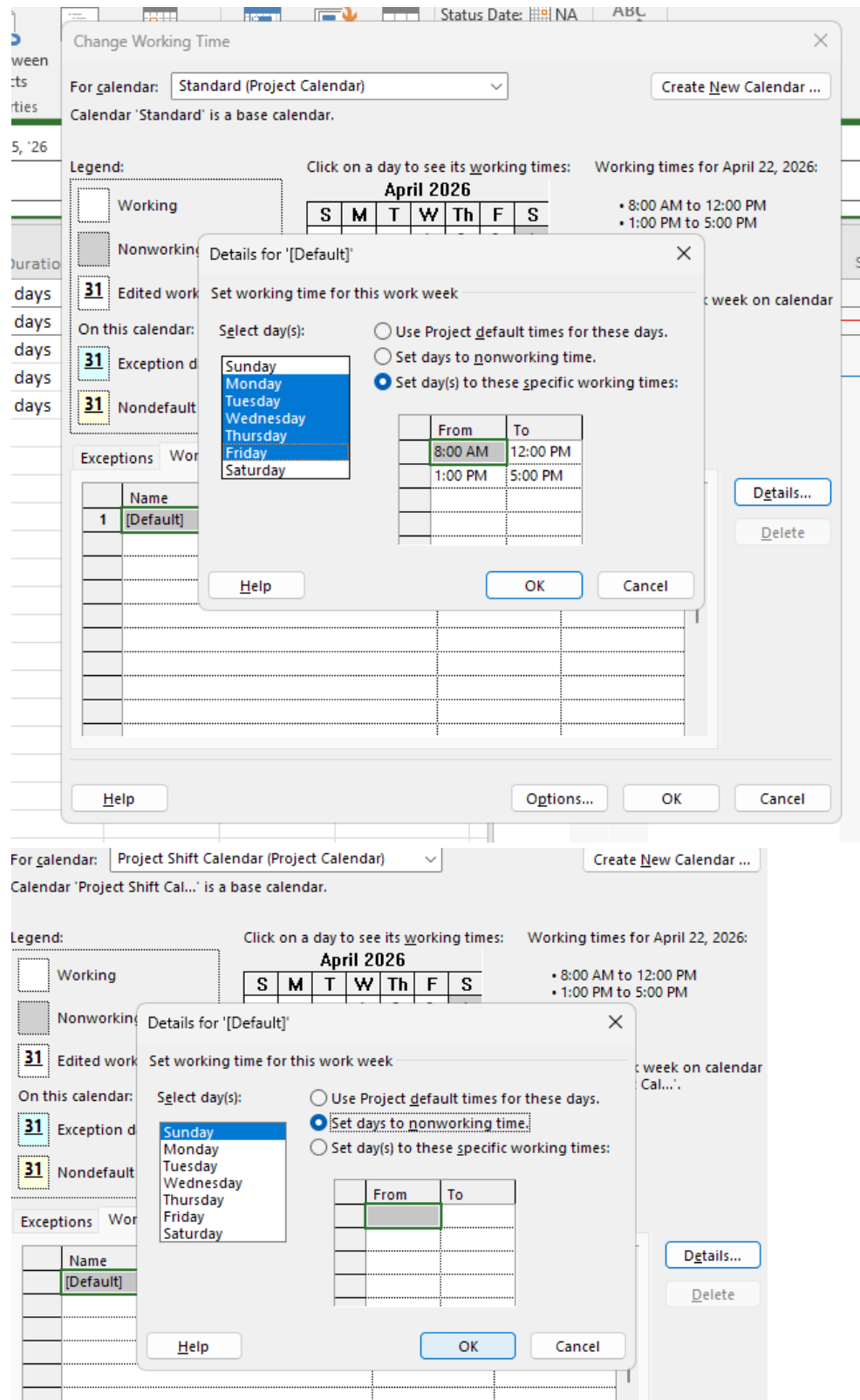
Step 1: Create Project Calendar

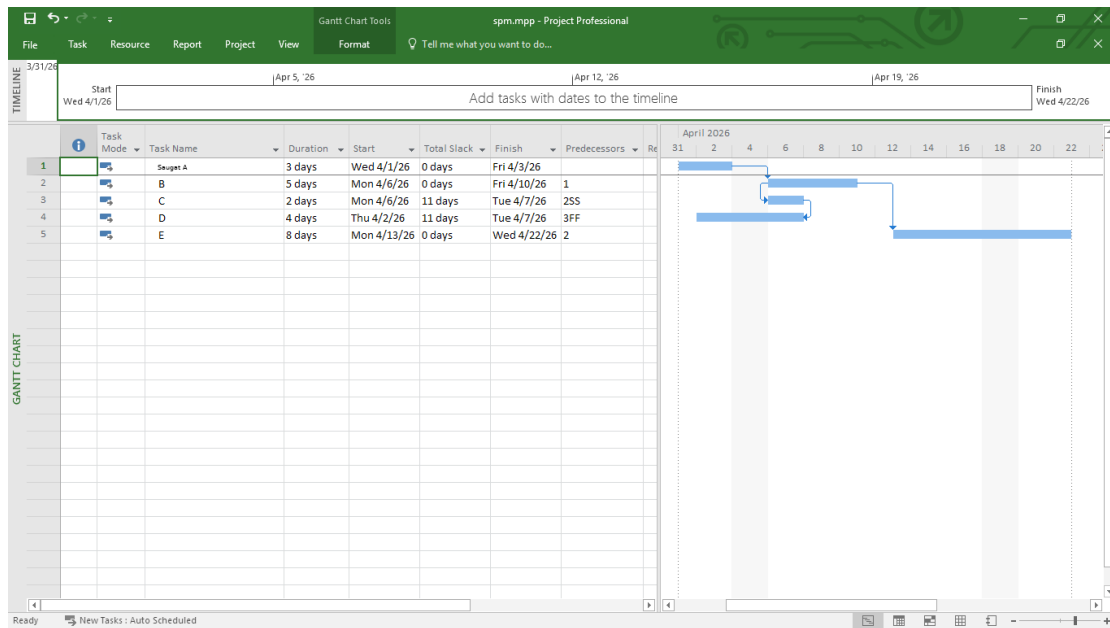
Opened the project management software and created a calendar for the project by setting working days and timing.



Step 2: Adjust Duty Shifts

Modified working hours and adjusted duty shifts for project activities and team members according to project requirements.





Result

A project calendar was successfully created, and duty shifts were adjusted according to project scheduling requirements.

Conclusion

Creating project calendars and adjusting duty shifts helps in effective time management, proper scheduling, and smooth project execution.

LAB NO. 4

Title: Evaluating Critical Path

Objective

- To identify and evaluate the critical path in a project schedule.
- To analyze task dependencies and project completion time.

Theory

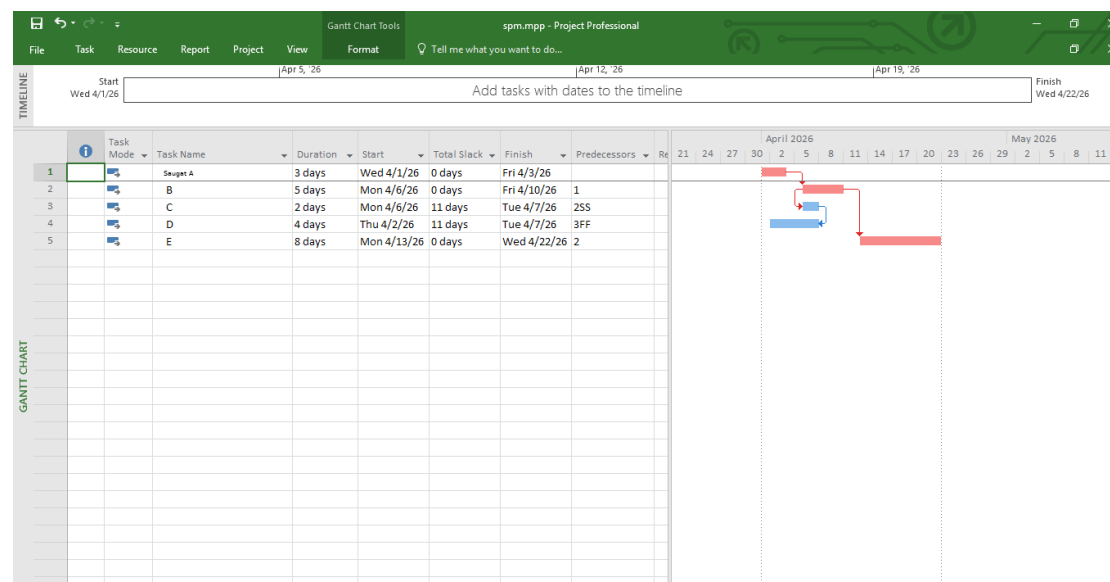
The critical path is the longest sequence of dependent tasks in a project that determines the minimum project completion time. Tasks on the critical path are called critical tasks because any delay in these tasks directly affects the overall project completion date.

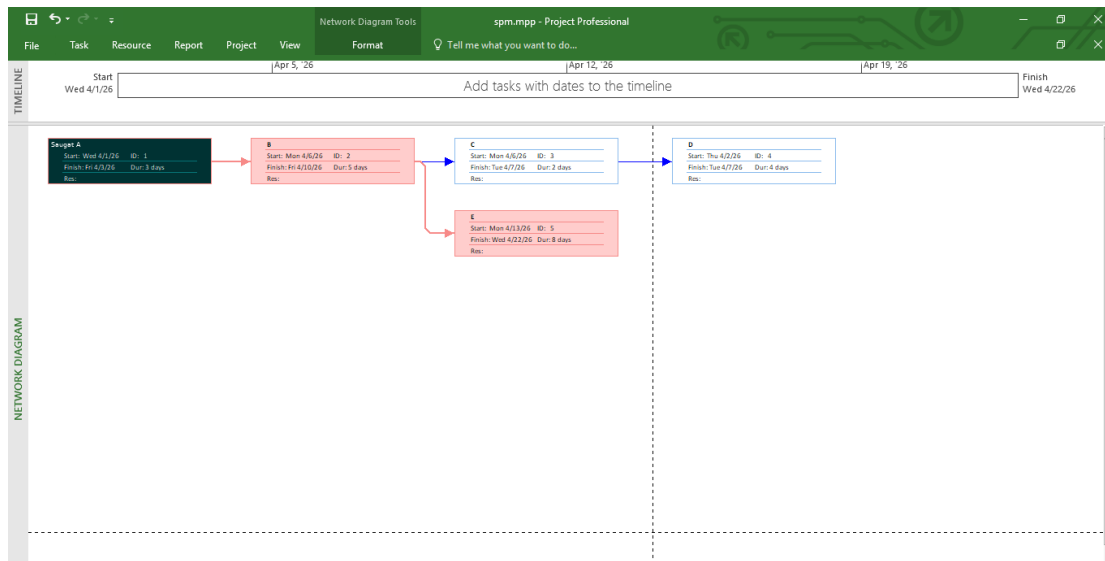
Critical Path Method (CPM) is used in project management to:

- Identify important tasks
- Calculate project duration
- Manage project scheduling effectively
- Reduce project delays

Procedure:

Used the Critical Path feature from the Project tab to identify critical tasks and analyze the project completion timeline in the Gantt Chart.





Result

The critical path of the project was successfully identified and evaluated using the project scheduling software. The software highlighted the critical tasks in the Gantt Chart, showing the sequence of activities that directly affect the project completion time. It was observed that any delay in these critical tasks would delay the entire project schedule.

Conclusion

Evaluating the critical path helps project managers identify the most important tasks in a project and monitor them carefully. It improves project planning, scheduling, and time management by focusing on tasks that have the greatest impact on project completion. The Critical Path Method also helps reduce delays and ensures successful completion of the project within the scheduled time.

LAB NO. 5

Title: Earned Value Analysis (EVA) : NPV, NFV and NAW

Objective

- To study Earned Value Analysis (EVA) in project management.
- To calculate Net Present Value (NPV), Net Future Value (NFV), and Net Annual Worth (NAW).
- To evaluate project cost and financial performance.

Theory

Earned Value Analysis (EVA) is a project management technique used to measure project performance in terms of cost and schedule. It helps project managers evaluate whether the project is progressing according to the planned budget and timeline.

Net Present Value (NPV)

NPV is the difference between the present value of cash inflows and outflows over a period of time. It is used to determine project profitability.

$$NPV = \sum_{t=1}^n \frac{R_t}{1+i} - C$$

Where:

- R_t = Net cash inflow at time period t
- i = Interest or discount rate
- t = Time period
- C = Initial investment cost

Net Future Value (NFV)

Net Future Value (NFV) represents the future value of the project investment after a certain period of time.

$$NFV = NPV (1 + i)^n$$

Where:

- NPV = Net Present Value
- i = Interest rate
- n = Number of years/time periods

Net Annual Worth (NAW) Net Annual Worth (NAW) converts the project value into equal annual amounts for easier comparison and evaluation.

$$NAW = NPV(A/P, i, n)$$

Where:

- NPV = Net Present Value
- A/P = Capital recovery factor
- i = Interest rate
- n = Number of years

These EVA techniques help organizations evaluate project feasibility, financial performance, and investment effectiveness.

Activities Performed

1. Collected project investment cost, annual cash inflow, interest rate, and project duration data.

Year	Cash Inflow (Rs.)	Discount Rate
0	-100000	10%
1	30000	10%
2	35000	10%
3	40000	10%
4	45000	10%
5	50000	10%

2. Calculated Net Present Value (NPV) using the EVA formula.

Year	Cash Inflow (Rs.)	Discount Rate
0	-100000	10%
1	30000	10%
2	35000	10%
3	40000	10%
4	45000	10%
5	50000	10%
NPV =	\$48,032.61	

3. Calculated Net Future Value (NFV) for future investment analysis.

Year	Cash Inflow (Rs.)	Discount Rate
0	-100000	10%
1	30000	10%
2	35000	10%
3	40000	10%
4	45000	10%
5	50000	10%
NPV =	\$48,032.61	
NFV =	\$77,357.00	

4. Calculated Net Annual Worth (NAW) for annual project evaluation.

Year	Cash Inflow (Rs.)	Discount Rate
0	-100000	10%
1	30000	10%
2	35000	10%
3	40000	10%
4	45000	10%
5	50000	10%
NPV =	\$48,032.61	
NFV =	\$77,357.00	
NAW =	\$12,670.88	

Result

NPV, NFV, and NAW values were successfully calculated using Earned Value Analysis techniques in spreadsheet software. The analysis showed that the project has a positive financial outcome and is economically beneficial. The calculated values helped evaluate the present profitability, future investment worth, and annual value of the project effectively.

Conclusion

Earned Value Analysis is an important project management technique used for evaluating project financial performance and investment feasibility. NPV helps determine present project profitability, NFV measures future investment value, and NAW converts project worth into equivalent annual values. These financial analysis methods support better project planning, budgeting, and decision-making for successful project management.

LAB NO. 6

Title: Software Effort Estimation using COCOMO

Theory

COCOMO (Constructive Cost Model) is a software cost estimation model developed by Barry Boehm for estimating the effort, development time, and manpower required for a software project. It helps project managers plan project resources, budget, and schedule before software development begins.

COCOMO estimates software effort based on the size of the project measured in KLOC (Kilo Lines of Code). The model is mainly divided into three project types:

- Organic
- Semi-Detached
- Embedded

The basic COCOMO model uses mathematical formulas to estimate software effort and development time.

Effort Estimation Formula

$$\text{Effort} = a * (\text{KLOC})^b$$

Where:

- Effort = Person-Months required
- KLOC = Thousand Lines of Code
- a and b = COCOMO constants

Development Time Formula

$$\text{Development Time} = c * (\text{Effort})^d$$

Where:

- Development Time = Time required to complete the project
- c and d = COCOMO constants
- Average Persons Required

$$\text{Persons} = \frac{\text{Effort}}{\text{Development Time}}$$

COCOMO helps organizations estimate project cost, manpower, and development duration effectively. It is widely used in software project planning and management.

Example Question

Estimate the effort, development time, and average number of persons required for a software project of size 20 KLOC using the Basic COCOMO Organic model.

Given:

$$\text{KLOC} = 20$$

$$a = 2.4$$

$$b = 1.05$$

$$c = 2.5$$

$$d = 0.38$$

Effort Calculation

$$\begin{aligned}\text{Effort} &= 2.4 * (20)^{1.05} \\ &= 55.17 \text{ Person-Months}\end{aligned}$$

Development Time Calculation

$$\begin{aligned}\text{Development Time} &= 2.5 * (55.17)^{0.38} \\ &= 11.46 \text{ Months}\end{aligned}$$

Average Persons Required

$$\begin{aligned}\text{Persons} &= 55.17 / 11.46 \\ \text{Persons Required} &= 4.81 \approx 5 \text{ Persons}\end{aligned}$$

Conclusion

COCOMO is an effective software cost estimation model used for calculating software development effort, project duration, and manpower requirements. It helps project managers estimate project resources and schedule accurately before development begins. Using the COCOMO model improves project planning, budgeting, and resource management for successful software development.

LAB NO. 7

Title: Risk Analysis and Risk Exposure Evaluation using Probability-Impact Matrix.

Objective

- To analyze project risks using risk exposure techniques.
- To evaluate risks using a probability-impact matrix.
- To identify risks above the tolerance line.
- To visualize risks using charts.

Theory

Risk analysis is the process of identifying, evaluating, and prioritizing project risks based on their probability and impact. Risk exposure is calculated by multiplying probability and impact scores. A probability-impact matrix helps classify risks according to their severity and assists project managers in making decisions.

Formula

$\text{Risk Exposure} = \text{Probability} \times \text{Impact}$

Given Risk Data

S.N.	Risk Name	Probability	Impact
1	Requirement Changes	High	Major
2	Key Staff Turnover	Medium	Critical
3	Budget Overrun	High	Major
4	Schedule Delay	High	Moderate
5	Technical Complexity	Medium	Major
6	Vendor Failure	Low	Major
7	Poor Communication	Medium	Moderate
8	Security Issues	Low	Critical
9	Hardware Failure	Medium	Minor
10	Regulatory Changes	Low	Major

1. Top 5 Risks Based on Risk Exposure

S.N.	Risk Name	Probability (0–1)	Impact (\$)	Risk Exposure
1	Requirement Changes	0.8	\$7,000.00	\$5,600.00
3	Budget Overrun	0.6	\$8,000.00	\$4,800.00
4	Schedule Delay	0.7	\$6,000.00	\$4,200.00
2	Key Staff Turnover	0.4	\$9,000.00	\$3,600.00
5	Technical Complexity	0.5	\$7,000.00	\$3,500.00
7	Poor Communication	0.6	\$5,000.00	\$3,000.00
10	Regulatory Changes	0.3	\$9,000.00	\$2,700.00
6	Vendor Failure	0.3	\$8,000.00	\$2,400.00
9	Hardware Failure	0.4	\$6,000.00	\$2,400.00
8	Security Issues	0.2	\$10,000.00	\$2,000.00

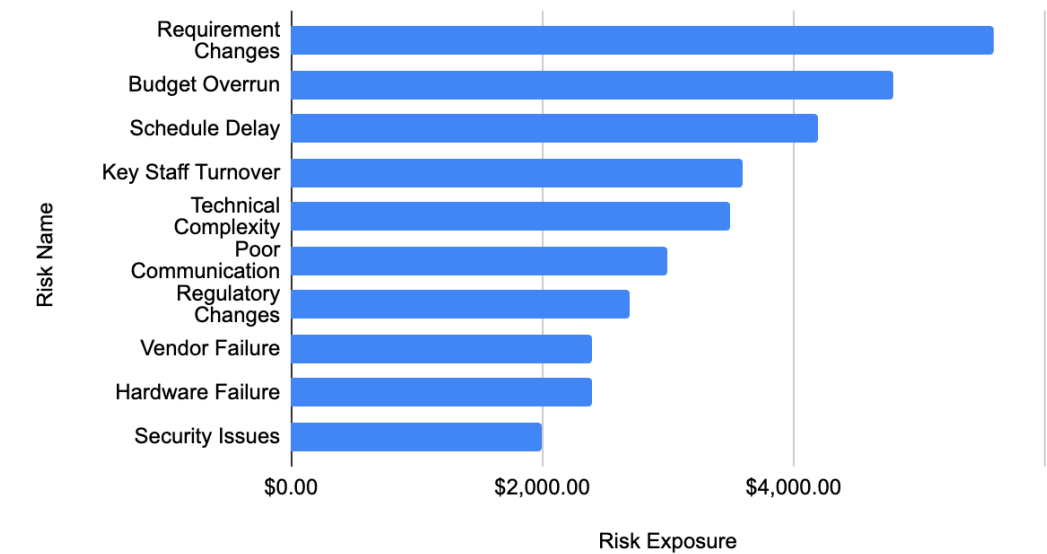
2. Probability-Impact Matrix Evaluation

	Critical		R8	R2	
	Major		R6,R10	R5	R1,R3
	Moderate			R7	R4
	Minor			R9	
		Very low	Low	Medium	High

3. Bar Chart / Pie Chart

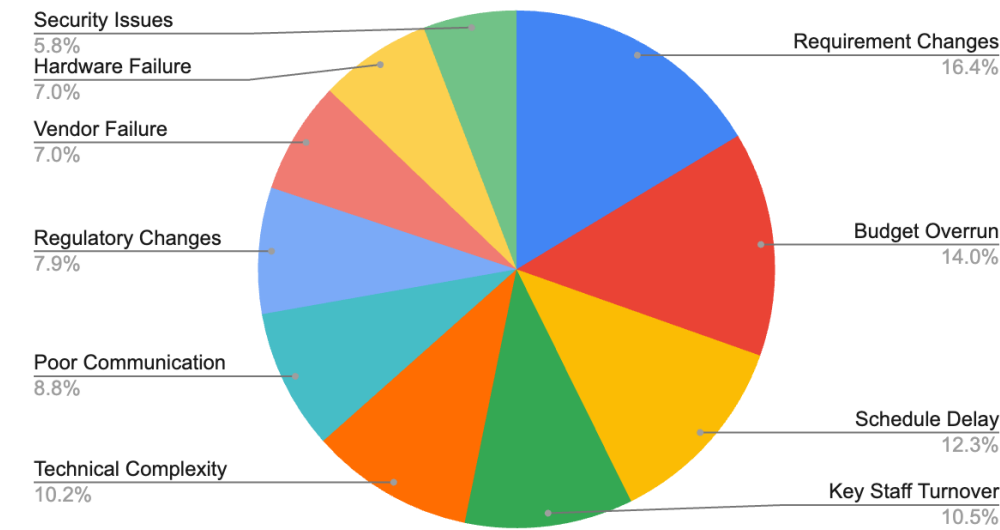
Bar Chart

Risk Exposure vs. Risk Name

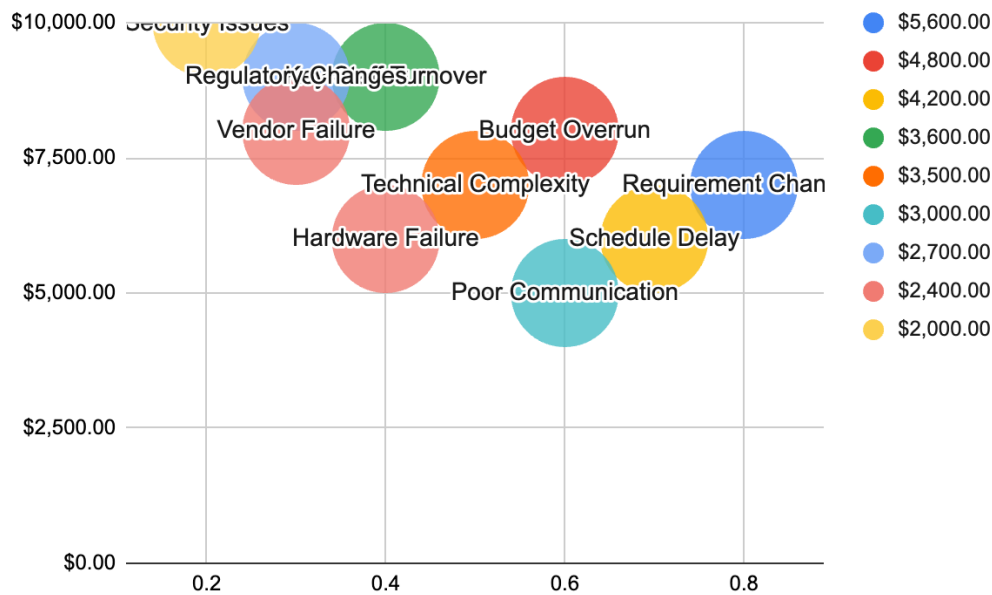


Pie Chart

Risk Exposure vs. Risk Name



4. Ball Chart (Bubble Chart)



Result

The project risks were successfully analyzed and evaluated using risk exposure calculation and probability-impact matrix techniques. The top risks were identified based on probability and impact values. Risks above the tolerance line were successfully finalized using the probability-impact matrix. Different visualization techniques such as bar chart, pie chart, and ball chart were created to represent project risks clearly and effectively.

Conclusion

Risk analysis is an essential process in project management that helps identify, evaluate, and control possible project risks. Risk exposure techniques and probability-impact matrix evaluation assist project managers in prioritizing critical risks and taking preventive actions. Visualization tools such as bar charts, pie charts, and bubble charts improve understanding of project risks and support better project planning and decision-making for successful project execution.